ARJUNA JEE 2027

Lecture -07 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A A.G.P.

B Properties of Sigma

C Question Practice

D





LAST CLASS RECAP

Answer the following Questions:

(i) Sum of n –AM's between a & b $= n\left(\frac{a+b}{2}\right) = n\left(\text{A.M. of } a+b\right)$

(ii) Product of n – GM's between a & b $= (ab)^{\frac{n}{2}} = \left(\sqrt{ab}\right)^{n} = \left(\text{G.M. of } a+b\right)^{n}$

(iii) sum of reciprocal of n–HM's between a & b $= \frac{n}{2}\left(\frac{1}{a}+\frac{1}{b}\right) = \frac{n}{\left(\text{H.M. of } a+b\right)}$

$= \frac{n}{\left(\frac{2}{\frac{1}{a}+\frac{1}{b}}\right)}$

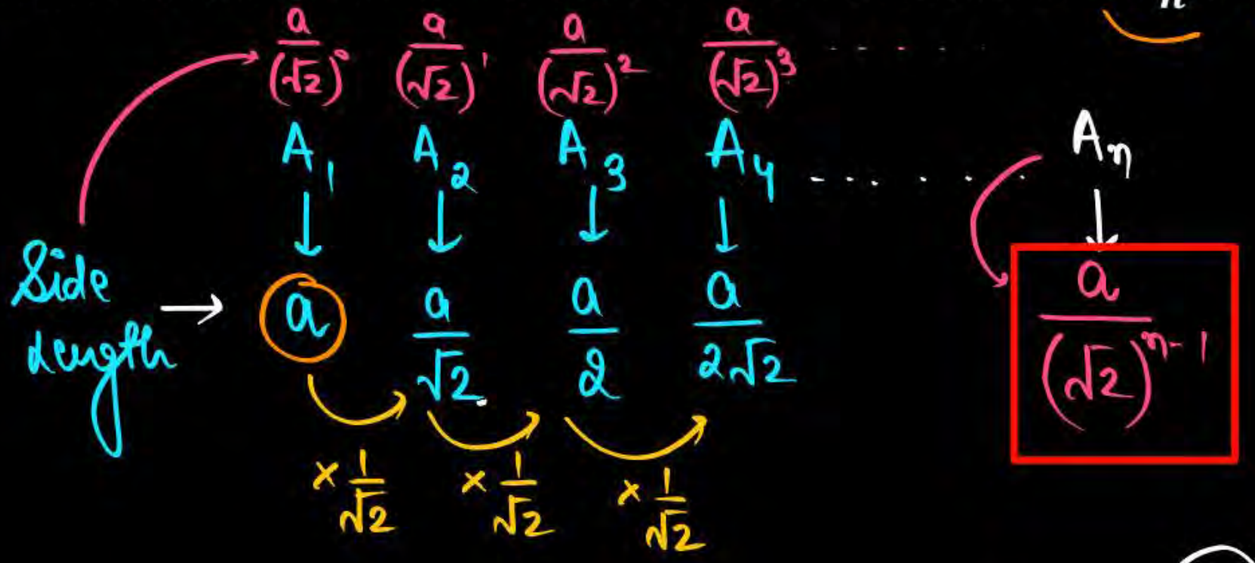
Question Practice

Question [JEE Mains-2021]



* $n=9$ fnf. 75%

Let $A_1, A_2, A_3, \ldots\ldots$ be squares such that for each $n \geq 1$, the length of the side of A_n equals the length of diagonal of A_{n+1}. If the length of A_1 is 12cm, then smallest value of n for which area of A_n is less than one, is

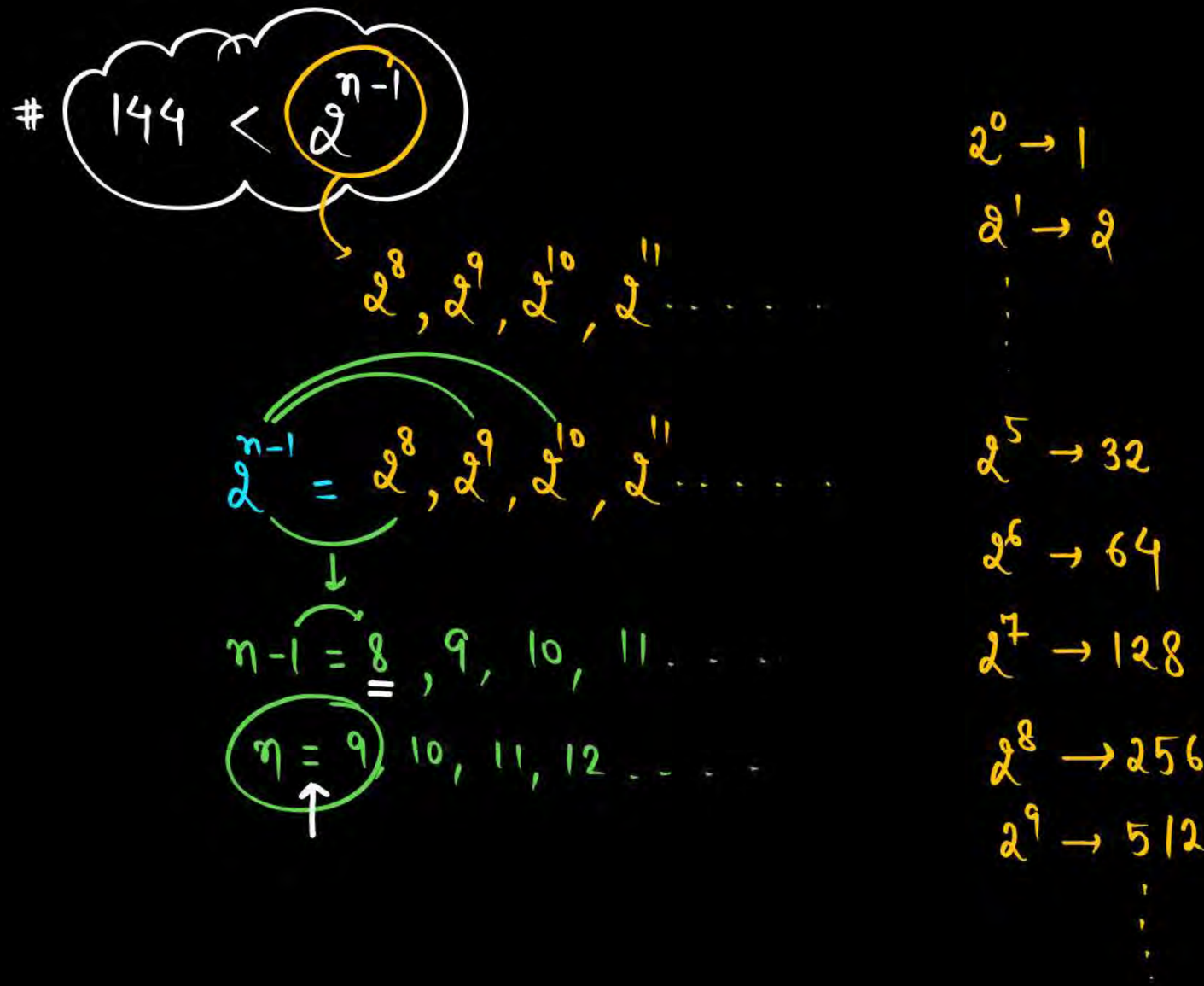


$a = 12\,cm$

$$\left((\sqrt{2})^{n-1}\right)^2 = \left((\sqrt{2})^2\right)^{n-1} = 2^{n-1}$$

$$\text{area of } A_n = \left(\frac{a}{(\sqrt{2})^{n-1}}\right)^2 < 1$$

$$\Rightarrow \frac{12^2}{2^{n-1}} < 1$$


144 < 2^{n-1}
$2^8, 2^9, 2^{10}, 2^{11}$ . . . . .
$2^{n-1} = 2^8, 2^9, 2^{10}, 2^{11}$ . . . . .
$n-1 = 8, 9, 10, 11$ . . .
$n = 9$ 10, 11, 12 . . . .
$2^0 \to 1$
$2^1 \to 2$
$2^5 \to 32$
$2^6 \to 64$
$2^7 \to 128$
$2^8 \to 256$
$2^9 \to 512$

Question

77%

If the pth, q^{th} and r^{th} terms of a Harmonic Progression (H.P.) are a, b and c respectively, then the value of $\frac{q-r}{a}+\frac{r-p}{b}+\frac{p-q}{c}$ is equal to.

A 1

B $p+q+r$

C 0

D abc

A.P. $\rightarrow$ $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ (p^{th}, q^{th}, r^{th} terms)

m, d

$T_p = m+(p-1)d = \frac{1}{a}$

$T_q = m+(q-1)d = \frac{1}{b}$

$T_r = m+(r-1)d = \frac{1}{c}$

$(p-r)d = \frac{1}{a}-\frac{1}{c}$ $\ominus$

$\left(\frac{r-p}{b}\right)d = \frac{1}{bc}-\frac{1}{ba}$

$\left(\frac{p-q}{c}\right)d = \left(\frac{1}{a}-\frac{1}{b}\right)\frac{1}{c} = \frac{1}{ac}-\frac{1}{bc}$

$\left(\frac{q-r}{a}\right)d = \left(\frac{1}{b}-\frac{1}{c}\right)\frac{1}{a} = \frac{1}{ab}-\frac{1}{ac}$

$\frac{1}{abc}-\frac{1}{aba}+\frac{1}{acc}-\frac{1}{bac}+\frac{1}{cab}-\frac{1}{acb} = 0$

Question [JEE Mains-2023]

Calculative

80%

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is 1 : 7 and $a + n = 33$, then the value of n is

A 21

B 22

C 23

D 24

$(n+2)$ terms

$a, A_1, A_2, \ldots, A_n, 100 \rightarrow$ A.P.

$a = 33 - n$

$T_{n+2} = a + (n+1)d = 100$

$(n+1)d = 100 - a$

$d = \frac{100-a}{n+1}$

$\frac{A_1}{A_n} = \frac{1}{7}$

$\frac{a+d}{a+nd} = \frac{1}{7}$

$7a + 7d = a + nd$

$6a = (n-7)d$

$6a = (n-7)\left(\frac{100-a}{n+1}\right)$

$6a = (n-7)\left(\frac{100-a}{n+1}\right)$

$a = 33 - n$

$6(33-n) = \frac{(n-7)(n+67)}{n+1}$

Santosh

$n = 21, 22, \checkmark 23, 24$

Satisfying only $n = 23$

Sumit:

$6(33-n)(n+1) = (n-7)(n+67)$

$6(33n - n^2 + 33 - n) = n^2 - 7n + 67n - 7\times 67$

$-6n^2 + 32\times 6n + 33\times 6 = n^2 + 60n - 7\times 67$

* PO-SHEN-LO.

$7n^2 - 132n - 469 - 198 = 0$

$7n^2 - 132n - 667 = 0$

23×29

Question [JEE Mains-2023]

"Simplification" (A.P + G.P)

Let A_1 and A_2 be two arithmetic means and G_1, G_2, G_3 be three geometric means of two distinct positive numbers. Then $G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$ is equal to

(A) $(A_1 + A_2)^2 G_1 G_3$

(B) $(A_1 + A_2) G_1^2 G_3^2$

(C) $2(A_1 + A_2) G_1^2 G_3^2$

(D) $2(A_1 + A_2) G_1 G_3$

$a, A_1, A_2, b \rightarrow$ A.P.

$a, G_1, G_2, G_3, b \rightarrow$ G.P. (r)

$ar \quad ar^2 \quad ar^3 \quad ar^4$

$G_1 G_3 = a^2 r^4 = a^2 \cdot \frac{b}{a} = ab$

$A_1 + A_2 = 2\left(\frac{a+b}{2}\right) = (a+b)$

$G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$

$a^4 r^4 + a^4 r^8 + a^4 r^{12} + a^4 r^8 = a^4 r^4 (1 + r^4 + r^8 + r^4)$

$= a^4 r^4 (r^8 + 2r^4 + 1)$

$a, A_1, A_2, b \rightarrow$ A.P.

$a, G_1, G_2, G_3, b \rightarrow$ G.P. (r)

$G_1 = ar, \quad G_2 = ar^2, \quad G_3 = ar^3, \quad b = ar^4$

$G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$

$= a^4r^4 + a^4r^8 + a^4r^{12} + a^4r^8$

$= a^4r^4\left(1 + r^4 + r^8 + r^4\right)$

$= a^4r^4\left(r^8 + 2r^4 + 1\right)$

$= a^3 \cdot \frac{b}{a}\left(\frac{b^2}{a^2} + 2\frac{b}{a} + 1\right)$

$= a^3 b\left(\frac{b^2 + 2ab + a^2}{a^2}\right)$

$= ab(a+b)^2 = G_1G_3(a+b)^2 = G_1G_3(A_1 + A_2)^2$

$b = ar^4$

$\# \quad \frac{b}{a} = r^4$

$\frac{b^2}{a^2} = r^8$

Question [JEE Mains-2015]

If m is the A.M. of two distinct real numbers l and $n(l, n > 1)$ and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals.



A $4l^2mn$

B $4lm^2n$

C $4lmn^2$

D $4l^2m^2n^2$



ARITHMETIC MEAN

Single A.M. : (Average)

\# A.M. of a & b = $\frac{a+b}{2}$

\# A.M. of a, b & c = $\frac{a+b+c}{3}$

\# A.M. of $a_1, a_2, a_3, \ldots\ldots, a_n$ = $\frac{a_1+a_2+a_3+\ldots+a_n}{n}$



GEOMETRIC MEAN

Single G.M. :

Geometric mean of a & b = $(ab)^{\frac{1}{2}} = \sqrt{ab}$

Geometric mean of a, b & c = $(abc)^{\frac{1}{3}}$

Geometric mean of $a_1, a_2, a_3, \ldots\ldots, a_n$ = $\left(a_1 \cdot a_2 \cdot a_3 \ldots a_n\right)^{\frac{1}{n}}$

HARMONIC MEAN

Single H.M. :

Harmonic mean of a & b = $\frac{2}{\left(\frac{1}{a}+\frac{1}{b}\right)}$

Harmonic mean of a, b & c = $\frac{3}{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)}$

Harmonic mean of $a_1, a_2, a_3 \ldots\ldots, a_n$ = $\frac{n}{\left(\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+\ldots\ldots+\frac{1}{a_n}\right)}$

• RAPID FIRE ROUND

1, 3, 4, 15

$$AM = \frac{1+3+4+15}{4} = \frac{23}{4}.$$

$$GM = (1\times3\times4\times15)^{\frac{1}{4}} = (180)^{\frac{1}{4}}$$

$$HM = \frac{4}{\left(\frac{1}{1}+\frac{1}{3}+\frac{1}{4}+\frac{1}{15}\right)} = \frac{240}{99} = \frac{80}{33}.$$

• RAPID FIRE ROUND

a, b, c → A.P. , middle term (b) is A.M. of a & c. ✓

a, b, c → G.P. , middle term (b) is G.M. of a & c. ✓

a, b, c → H.P. , middle term (b) is H.M. of a & c. ✓

$2b = a + c$

$b = \frac{a+c}{2}$

$b^2 = ac$

$b = \pm\sqrt{ac}$

$b = \sqrt{ac} = (ac)^{\frac{1}{2}}$.

$b = \frac{2ac}{a+c} = \frac{2}{\left(\frac{1}{a} + \frac{1}{c}\right)}$

Ques:

$-1, 3, 4, 15$

$G.M. = (-1 \times 3 \times 4 \times 15)^{\frac{1}{4}} \rightarrow$ n.d.

$-2, 1, 4$

$GM = (-8)^{\frac{1}{3}} = -2.$

$-2, 2$

$GM = (-4)^{\frac{1}{2}} \rightarrow$ n.d.

$= \sqrt{-4}$

Quad. whose roots are a & b :-

$AM = \frac{a+b}{2}$ $\quad$ $GM = \sqrt{ab}$

$x^2 - (a+b)x + ab = 0$

$x^2 - 2(AM)x + (GM)^2 = 0$

Cubic whose Roots are a, b, c :

$$AM = \frac{a+b+c}{3}$$

$$GM = (abc)^{\frac{1}{3}}$$

$$HM = \frac{3}{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)} = \frac{3}{\left(\frac{bc+ac+ab}{abc}\right)}$$

$$x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc = 0 .$$

$$\boxed{x^3 - (3\,A.M.)x^2 + \left(\frac{3(G.M.)^3}{H.M.}\right)x - (GM)^3 = 0 .}$$

$$HM = \frac{3abc}{ab+bc+ca} .$$

$$ab+bc+ca = \frac{3(GM)^3}{HM}$$

Ques:

Find n if $\left(\frac{a^{n+1}+b^{n+1}}{a^n+b^n}\right)$ is G.M. of a & b.

Sol^n

$\left(\frac{a^{n+1}+b^{n+1}}{a^n+b^n}\right) = (ab)^{\frac{1}{2}} = a^{\frac{1}{2}} b^{\frac{1}{2}}$

$a = b$

$\frac{2a^{n+1}}{2a^n} = a$

$a^{n+1} + b^{n+1} = a^{\frac{1}{2}} b^{\frac{1}{2}} a^n + a^{\frac{1}{2}} b^{\frac{1}{2}} b^n$

$a^{n+1} - a^{\frac{1}{2}} b^{\frac{1}{2}} a^n = a^{\frac{1}{2}} b^{\frac{1}{2}} b^n - b^n \cdot b^{\cdot}$

$a^n \cdot a^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}}\right) = b^n b^{\frac{1}{2}} \left(a^{\frac{1}{2}} - b^{\frac{1}{2}}\right)$

$a^{n+1} - a^{\frac{1}{2}} \cdot b^{\frac{1}{2}} \cdot a^n$

$a^n \cdot a^1 - a^{\frac{1}{2}} \cdot b^{\frac{1}{2}} \cdot a^n$

$a^n \; a^{\frac{1}{2}} \cdot a^{\frac{1}{2}} - a^{\frac{1}{2}} b^{\frac{1}{2}} a^n$

$a^{n+\frac{1}{2}} = b^{n+\frac{1}{2}}$

$n + \frac{1}{2} = 0 \rightarrow n = -\frac{1}{2}$



Very Important Result

For two distinct positive numbers a & b :

$$\left(\frac{a^{n+1}+b^{n+1}}{a^n+b^n}\right)$$

If $n = 0$ → AM of a & b.

If $n = \frac{-1}{2}$ → GM of a & b.

If $n = -1$ → HM of a & b.

Example

The A.M. between two positive numbers exceeds the G.M. by 5, and the G.M. exceeds the H.M. by 4. Find the numbers.



A.G.P. (ARITHMETIC GEOMETRIC PROGRESSION)

$$a,\ (a+d)r,\ (a+2d)r^2,\ (a+3d)r^3,\ (a+4d)r^4 \ldots \rightarrow \text{A.G.P.}$$

(each step $\times r$)

How to find Sum

(Multiply by r, shift & subtract.

Above method is also used when successive terms are (×) by r.

Question [JEE Mains-2019] (1)

$$S = \sum_{k=1}^{\infty} \frac{k}{2^k}$$

$$\Rightarrow S = \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \ldots\ldots\ldots\infty$$

$\frac{1}{2} \times$

$$\frac{S}{2} = 0 + \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \frac{4}{2^5} + \ldots\ldots\infty$$

$\ominus$

$$S - \frac{S}{2} = \boxed{\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots\ldots\infty} \rightarrow \text{G.P.}$$

$$\frac{S}{2} = \frac{\left(\frac{1}{2}\right)}{1 - \frac{1}{2}} = 1 \rightarrow \frac{S}{2} = 1 \rightarrow \boxed{S = 2}$$ fns

Example ②

Find sum of "n-terms" & S_∞ : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$

$\times\frac{1}{3}$ $\times\frac{1}{3}$ $\times\frac{1}{3}$

$$S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots + T_{n-1} + T_n$$

$\frac{1}{3}\times$

$$\frac{S}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots\ldots \qquad \frac{T_{n-1}}{3} + \frac{T_n}{3}$$

$\ominus$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\ldots + \left(T_n - \frac{T_{n-1}}{3}\right) - \frac{T_n}{3}$$

$$S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots + T_{n-1} + T_n$$

$\times \frac{1}{3}$

$$\frac{S}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots\ldots + \frac{T_{n-1}}{3} + \frac{T_n}{3}$$

$\ominus$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\ldots + \left(T_n - \frac{T_{n-1}}{3}\right) - \frac{T_n}{3}$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\ldots\ldots\ldots\right) - \frac{T_n}{3}$$

$\Rightarrow$ G.P. $\quad a = 1, \; r = \frac{1}{3}$

n.o.t. $= n-1$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots\dots\dots\dots\dots\right) - \frac{T_n}{3}$$

G.P. $a = 1,\ r = \frac{1}{3},\ n.o.t. = n-1$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(\frac{1\left(1-\left(\frac{1}{3}\right)^{n-1}\right)}{\frac{2}{3}}\right) - \frac{1}{3}\left(T_n\right)$$

Pattern observation

$$\frac{2S}{3} = \frac{3}{5} + \frac{1}{5}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{3}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

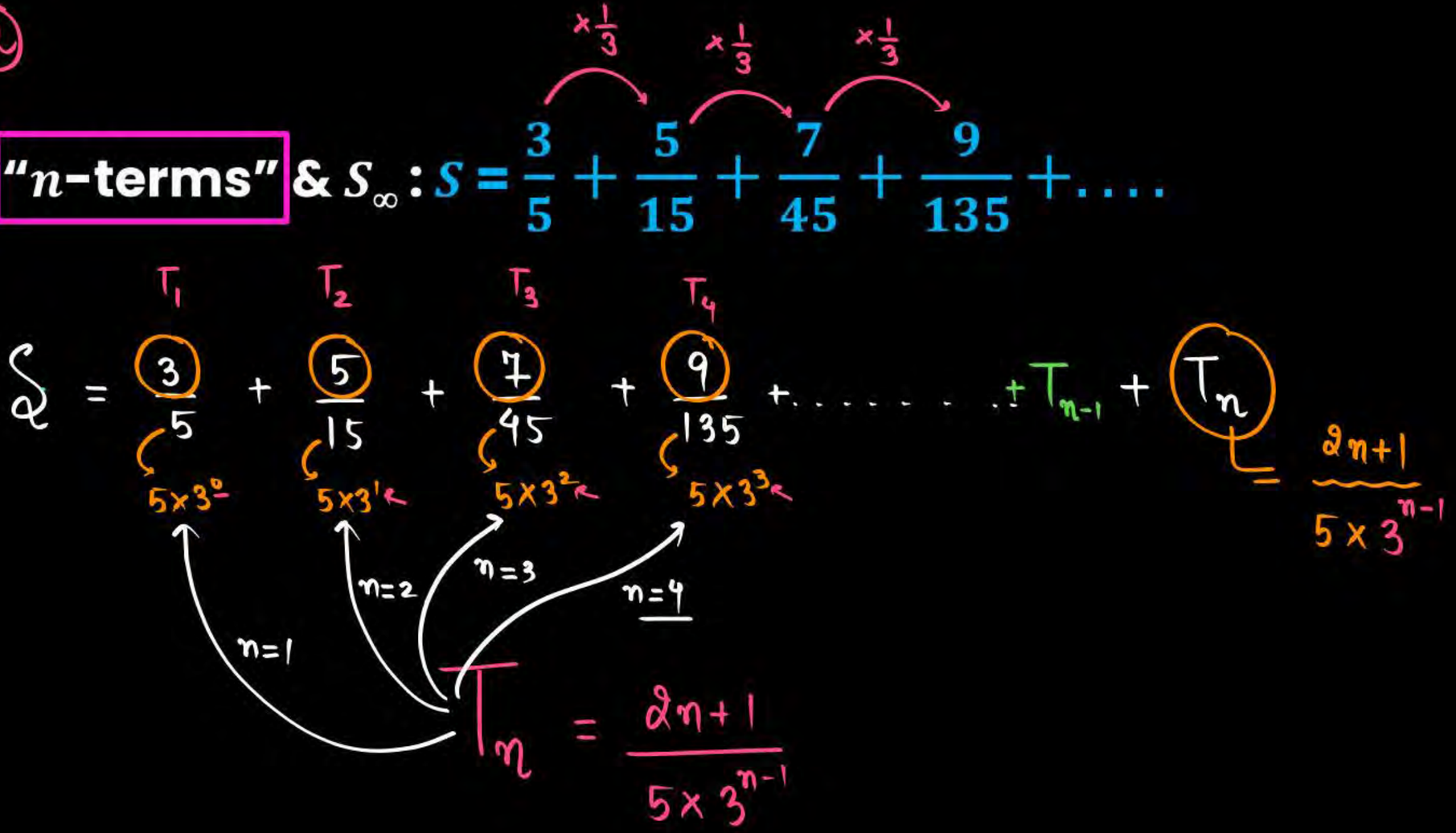
$\times \frac{3}{2}$

$$S = \frac{9}{10} + \frac{3}{10}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{2}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

→ Sum of n-Terms

Example (2)

Find sum of "n-terms" & S_∞ : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$



Example ②

Find sum of "n-terms" & S_∞ : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$ ($\times\frac{1}{3}$, $\times\frac{1}{3}$, $\times\frac{1}{3}$)

$$S_\infty = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots\ldots\ldots\ldots\ldots\infty$$

$\frac{1}{3}\times$

$$\frac{S_\infty}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots\ldots\ldots\ldots\ldots\infty$$

$\ominus$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\ldots\ldots\ldots\ldots\infty$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\infty\right)$$

$\frac{1}{1-\frac{1}{3}} = \frac{3}{2}$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\times\frac{3}{2}$$

$$\frac{2S}{3} = \frac{4}{5}$$

$$S_\infty = \frac{6}{5}$$

Question [JEE Mains-2023]

If the sum of the series

$$\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot 3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2\cdot 3^3}+\frac{1}{3^4}\right)+\cdots$$

is $\frac{\alpha}{\beta}$, where α and β are co-prime, then $\alpha+3\beta$ is equal to



Challenger Question [JEE Mains-2021]

Consider an arithmetic series and a geometric series having four initial terms from the set {11, 8, 21, 16, 26, 32, 4}. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to



DO IT BY YOURSELF (DIBY)

GEOMETRIC PROGRESSION

DIBY-34

Let $a_1, a_2, a_3, \ldots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to [JEE Mains-2025]

(A) 131 (B) 130 (C) 129 (D) 128

[Ans. : (C)]

DIBY-35

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to: [JEE Mains-2025]

(A) 812 (B) 526 (C) 784 (D) 628

[Ans. : (C)]

DIBY-36

Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b, is equal to p^{th} term of another GP, whose first term is a and fifth term is b. Then p is equal to [JEE Mains-2024]

(A) 20 (B) 25 (C) 21 (D) 24

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-37

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be $\frac{1}{9}$.

Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to [JEE Mains-2023]

(A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{3}$ (D) 3

[Ans. : (D)]

DIBY-38

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th terms is: [JEE Mains-2024]

(A) 96 (B) 78 (C) 91 (D) 84

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-39

Find the four successive terms of a. G.P. of which the 2nd term is smaller than the first by 35 and the 3rd term is larger than the 4th by 560.

(A) $7, -28, 112, -448$

(B) $5, -20, 80, -320$

(C) $7, 28, 112, 448$

(D) $-\frac{35}{3}, -\frac{140}{3}, -\frac{560}{3}, -\frac{2240}{3}$

[Ans. : A, D]

DIBY-40

If a_1, a_2, a_3 are in G.P. such that $a_1 + a_2 + a_3 = 13$ and $a_1^2 + a_2^2 + a_3^2 = 91$. Find a and r.

(A) $(a = 1, r = 2)$ or $\left(a = 4, r = \frac{1}{2}\right)$

(B) $(a = 3, r = 1)$ or $\left(a = \frac{1}{3}, r = 9\right)$

(C) $(a = 2, r = 3)$ or $\left(a = 9, r = \frac{1}{2}\right)$

(D) $(a = 1, r = 3)$ or $\left(a = 9, r = \frac{1}{3}\right)$

[Ans. : (D)]

GEOMETRIC PROGRESSION

DIBY-41

If $S_1, S_2, S_3, \ldots . S_p$ are the sums of infinite G.P. whose first terms are $1, 2, 3, \ldots \ldots p$, and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \ldots \frac{1}{p+1}$ respectively, then find the value of $S_1 + S_2 + S_3 + \cdots S_p$.

(A) $\frac{p(p+1)}{2}$ (B) $\frac{(p+1)(p+3)}{2}$ (C) $\frac{p}{2}(p+3)$ (D) $p(p+3)$

[Ans. : (C)]

DIBY-42

The sum of an infinite number of terms of a G.P. is 15 and the sum of their squares is 45. Find the series.

(A) $5, \frac{10}{3}, \frac{20}{9}, \frac{40}{27}, \ldots$ (B) $3, 2, \frac{4}{3}, \frac{8}{9}, \ldots$

(C) $5, \frac{5}{3}, \frac{5}{9}, \frac{5}{27}, \ldots$ (D) $3, \frac{6}{5}, \frac{12}{25}, \ldots$

[Ans. : (A)]

GEOMETRIC PROGRESSION

DIBY-43

If the sum of an infinite GP a, ar, ar^2, ar^3, the squares of its each term is 150, then the sum of ar^2, ar^4, ar^6, is: [JEE Mains-2021]

(A) $\frac{1}{2}$ (B) $\frac{25}{2}$ (C) $\frac{5}{2}$ (D) $\frac{9}{2}$

[Ans. : (A)]

DIBY-44

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to: [JEE Mains-2021]

(A) $7 + \sqrt{3}$ (B) $7 - \sqrt{3}$ (C) $7 + 3\sqrt{3}$ (D) $7 - 7\sqrt{3}$

[Ans. : (A)]

DIBY-45

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is: [JEE Mains-2020]

(A) $\frac{2}{13}(3^{50}-1)$ (B) $\frac{1}{13}(3^{50}-1)$

(C) $\frac{1}{26}(3^{49}-1)$ (D) $\frac{1}{26}(3^{50}-1)$ [Ans. : (D)]

DIBY-46

The greatest positive integer k, for which 49^k+1 is a factor of the sum $49^{125}+49^{124}+\cdots+49^2+49+1$, is [JEE Mains-2020]

(A) 32 (B) 63 (C) 65 (D) 60

[Ans. : (B)]

DIBY-47

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is: [JEE Mains-2025]

(A) 760 (B) 755 (C) 750 (D) 757

[Ans. : (D)]

DIBY-48

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? [JEE Mains-2019]

(A) d, e, f are in A.P.

(B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(D) d, e, f are in G.P.

[Ans. : (C)]

DIBY-49

Find the sum of the following series

(i) 5 + 55 + 555 + ... to n terms

(ii) 0.3 + 0.33 + 0.333 + ... to n terms

[Ans. : (i) $\frac{5}{9}\left[10\left(\frac{10^n-1}{9}\right)n\right]$ (ii) $\frac{1}{3}\left[n-\frac{1}{9}\left(1-\frac{1}{10^n}\right)\right]$]

DIBY-50

If $x = 1 + a + a^2 + a^3 + \ldots\ldots \infty$ and $y = 1 + b + b^2 + b^3 + \ldots\ldots \infty$ show that

$1 + ab + a^2b^2 + a^3b^3 + \ldots\ldots \infty = \frac{xy}{x+y-1}$ where $0 < a < 1$ and $0 < b < 1$.

It is never about end result

It's always about Journey

Thank You

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